

COLLISION REPAIR INFORMATION BULLETIN



SUBJECT:	STRUCTURAL REPAIR PRECAUTIONS		
BULLETIN NO.:	TS-TCI-CRIB-2036	DATE ISSUED:	August, 2013
MODELS:	All Toyota & Scion Models	SECTION:	Structural

INTRODUCTION:

Changes in the structural configuration of steel can take place when it is heated during repairs. Generally, when steel is heated enough, its tensile strength can decrease. When steel is heated above a certain fixed temperature (about 399° C or 750° F), the qualities of the metal will change. Using heat to stress relieve structural steel can result in:

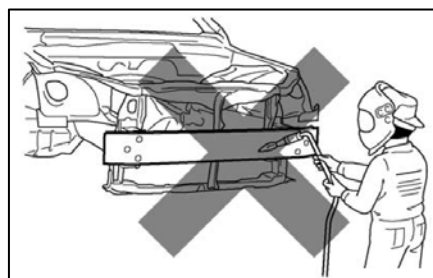
- Destruction of corrosion resistance coatings
- Decreased impact energy absorption capacity
- Brittleness of the metal
- Cracking of the metal

Three types of High Strength Steel (H.S.S.) are used in the construction of Toyota/Scion vehicles:

- Solution Hardened H.S.S. typically used for outer body panels, suspension-related components, unibody structural elements and perimeter frames.
- Precipitation Hardened Steel which is a high-tensile strength steel used mainly for door intrusion beams and bumper reinforcements.
- Dual Phase Steel referred to as quenched martensitic and ferritic structure which is also used for door intrusion beams and bumper reinforcements.

PRECAUTIONS:

Toyota/Scion does not approve of the use of open flame heating to stress relieve High Strength Steel (H.S.S.) because it is extremely difficult to accurately control the temperature at the correct level in actual repair operations, which subsequently deteriorates the quality, durability and strength of the steel.



NOTE: Issue of this Collision Repair Information Bulletin by Toyota Canada Inc. is to ensure that proper repair information is available. It does not constitute an authorization for repair. PLEASE ROUTE THIS BULLETIN TO YOUR COLLISION REPAIR CENTER MANAGER AND COLLISION & REFINISH TECHNICIANS.



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REPAIR INFORMATION:

“Cold Straightening” (stress relieving) is the only repair method use to repair unibody structural members and perimeter frames (excludes all components noted in “Replacement Information”). If the structural member or perimeter frame cannot be satisfactorily restored to original dimensions during this process, it should be replaced.

REPLACEMENT INFORMATION:

Replacement and sectioning procedures, as well as location of H.S.S. structural members and body panels are located in each model-specific Repair Manual for Collision Damage.

For perimeter frame vehicles, no sectioning is recommended; therefore, the complete frame must be replaced.

The following H.S.S. components must always be replaced when damaged:

- Bumper reinforcements
- Suspension components
- Door intrusion beams (only available with door shell assembly)

Toyota and Scion only support the use of OEM replacement parts for collision repair.

Refer to the appropriate Toyota/Scion Collision Repair Manuals for more information. For Toyota or Scion Collision Centre facilities, visit the ETAS and Toyota Technical Information System (TIS) site located in the Infostream website portal at <http://infostream.toyota.ca/> to access Repair Manuals for Collision Damage and other Collision Repair Information Bulletins. For independently operated collision centres interested in these publications, visit www.techinfo.toyota.com for a subscription and access.